

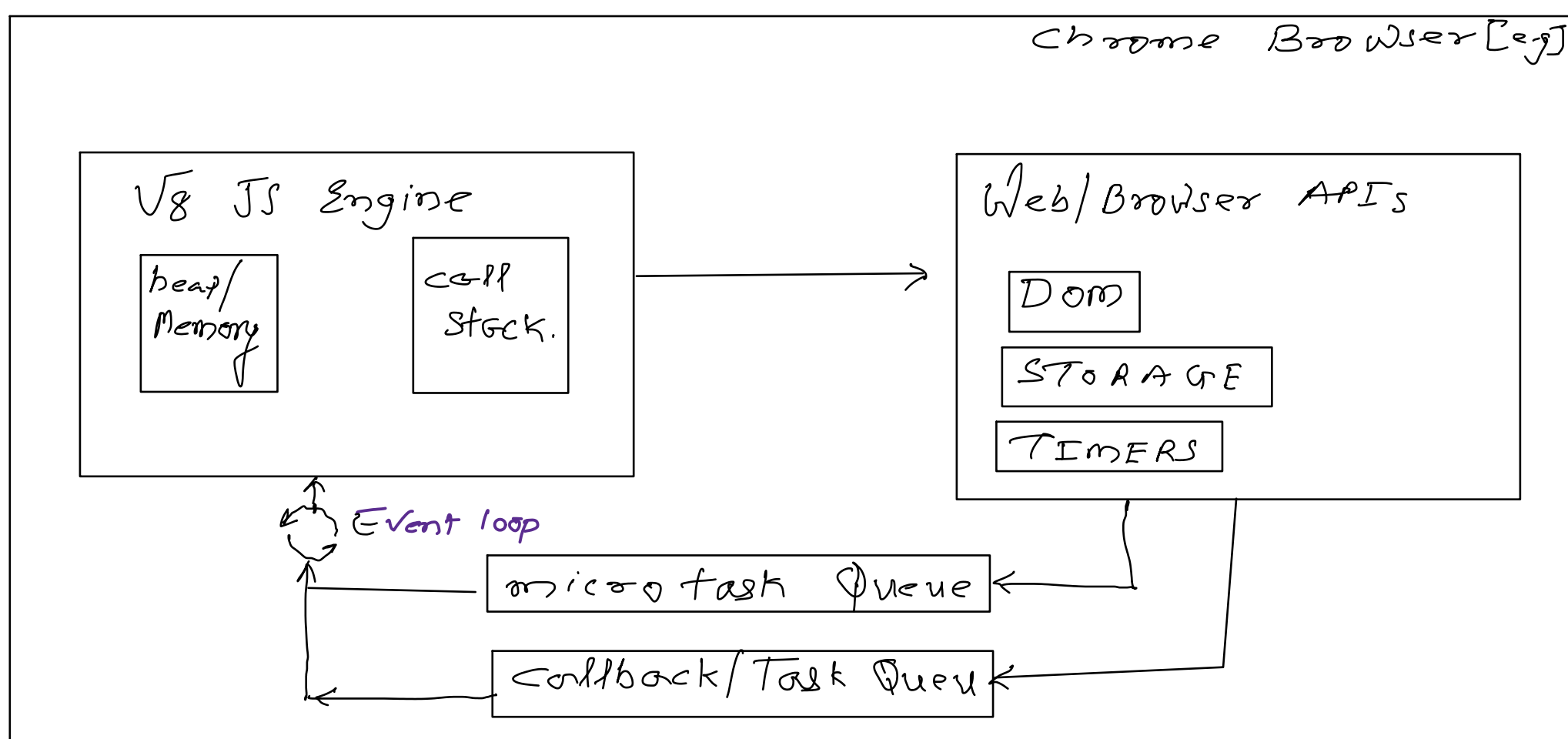
0004 JavaScript RunTime

Tuesday, 20 June 2023 9:00 AM

- JavaScript runtime is an environment which provides all the necessary components in order to use & run a JavaScript program.
- Every Browser has a JavaScript Engine.
- A JavaScript Engine is one component in the JavaScript Runtime.
- So Q. is what else JS Runtime consist of?

e.g.:-

JavaScript Code



- JS engine is V8 engine which executes JS code
- JS engine consist Call Stack - Where JS code can executed & A Heap is nothing but memory where All variable stores that JS Application needs.
- Web API Such as **DOM**
Timers - such as set time out & Time interval, Promises, Browser Storages etc which add extra functionality to JS engine but are not part of JS languages itself.
- All above Web APIs Provided by browser itself.
- Queues are where Asynchronous task read before they can be executed.
- Event loop, make insures Async task must be executed in right order.
- Above is Very high level introduction to JS Runtime with Async task.
- In short JS Engine is only Not sufficient in this JS Runtime. Engine itself can execute ECMA Script but JS Script = ECMA Script + All above components of JS Runtime.
- Concept So far:-
 - ECMA Script
 - JavaScript Engine
 - JavaScript Runtime.